CMAI 480

Computer Vision

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PROJECTION GEOMETRY

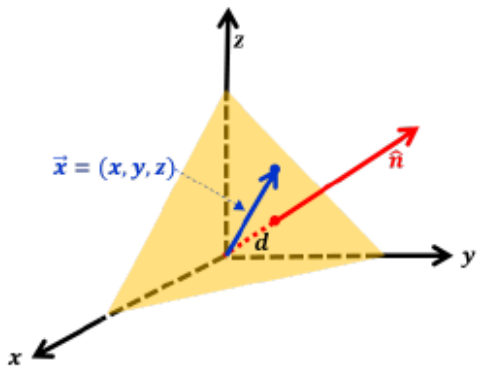
HOMOGENEOUS REPRESENTATION OF WORLD 3D ENTITIES: POINTS, PLANES, LINES AND QUADRICS

Points in 3D

- We represent a point in 3D physical space $\mathbb{R}^3$ by $(X,Y,Z)^T$.
- Its homogeneous coordinates in $\mathbb{R}^4 - (0,0,0,0)^T$ will be denoted by the vector $(x_1,x_2,x_3,x_4)^T$ with $X = x_1/x_4$, $Y = x_2/x_4$, $Z = x_3/x_4$.
  - The notation $-(0,0,0,0)^T$ indicates that the vector $(0,0,0,0)^T$, which doesn't correspond to any point/plane, is excluded.
- All points $k.(x_1,x_2,x_3,x_4)^T$ for all non-zero k form an equivalence class in $\mathbb{R}^4$, in the sense that they all represent the same physical point in $\mathbb{R}^3$.
- Using the same arguments as in Lecture 2, a point at infinity in physical $\mathbb{R}^3$ is represented by homogeneous coordinates of type $(x_1,x_2,x_3,0)^T$.
- Using homogeneous coordinates, a transformation of the physical space $\mathbb{R}^3$ can be represented by a 4×4 homography: $X' = HX$. Since only the ratios matter, in general the homography H has 15 degrees of freedom.

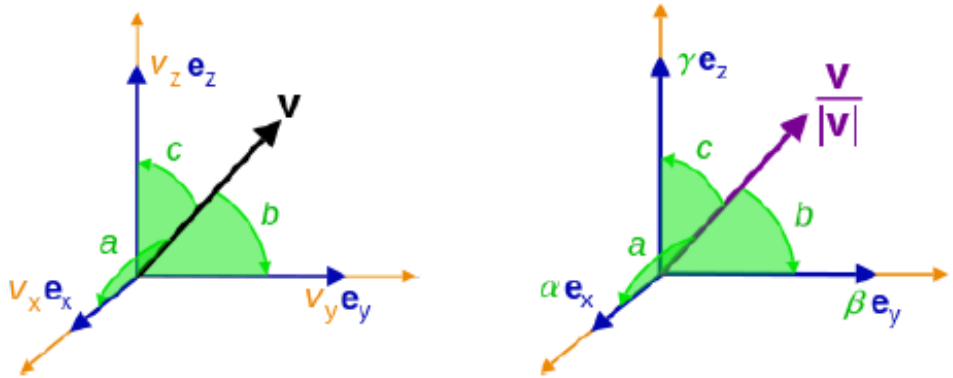
Planes in 3D

- Perhaps the most common way to represent a plane is through the geometrical construction at below.



- The plane is characterized by a unit normal vector $\hat{n}$ passing through the origin and the shortest distance d to the plane from the origin (This is along $\hat{n}$).
- Now given any point $\vec{X} = (X, Y, Z)^T$ on the plane, it follows straight forwardly that $\vec{X}^T . \hat{n} = d$.
- This equation can be written as $n_x X + n_y Y + n_z Z - d = 0$ where n_x, n_y, n_z are the direction cosines associated with the unit normal $\hat{n}$.
  - Direction cosine refers to the cosine of the angle between any two vectors.

Normal Vector as Cosine Directions



If v is an Euclidean vector in three-dimensional Euclidean space, $\mathbb{R}^3$, $v = v_x e_x + v_y e_y + v_z e_z$, where e_x, e_y, e_z are the standard basis in Cartesian notation, then the direction cosines are

$$\alpha = \cos a = \frac{v \cdot e_x}{|v|} = \frac{v_x}{\sqrt{v_x^2 + v_y^2 + v_z^2}}, \tag{1}$$

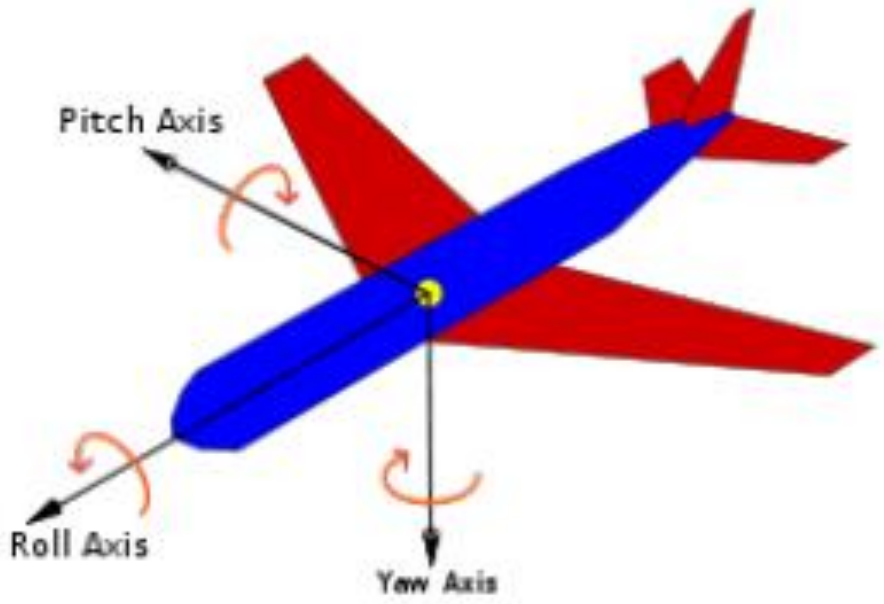
$$\beta = \cos b = \frac{v \cdot e_y}{|v|} = \frac{v_y}{\sqrt{v_x^2 + v_y^2 + v_z^2}}, \tag{2}$$

$$\gamma = \cos c = \frac{v \cdot e_z}{|v|} = \frac{v_z}{\sqrt{v_x^2 + v_y^2 + v_z^2}}. \tag{3}$$

It follows that by squaring each equation and adding the results:
$\cos^2 a + \cos^2 b + \cos^2 c = 1$. Here, α, β and γ are the direction cosines and the Cartesian coordinates of the unit vector $v/|v|$, a, b and c are the direction angles of the vector v.

Euler Angles

- Refer to the textbook or Wikipedia
  - http://en.wikipedia.org/wiki/Euler_angles

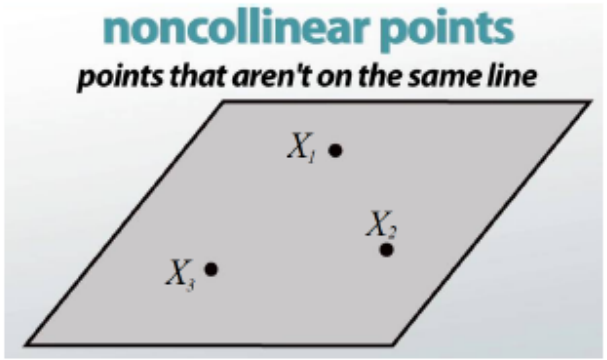


Representing a Plane in 3D

- Therefore, we can write $\pi_1 X + \pi_2 Y + \pi_3 Z + \pi_4 = 0$ as a general representation of a plane through a 4-vector parameter $(\pi_1, \pi_2, \pi_3, \pi_4)^T$.
  - Using homogeneous coordinates, this translates into $\pi_1 x_1 + \pi_2 x_2 + \pi_3 x_3 + \pi_4 x_4 = 0$.
- We thus have the following compact representation of a plane in homogeneous coordinates $\Pi^T X = 0$ or $X^T \Pi = 0$
  - where $\Pi = (\pi_1, \pi_2, \pi_3, \pi_4)^T$ and $X = (x_1, x_2, x_3, x_4)^T$.
- We also recognize that the unit normal to the plane is encoded in (π_1, π_2, π_3) and the perpendicular distance to the plane in π_4.
- More precisely (π_1, π_2, π_3) is the normal to the plane in physical $\mathbb{R}^3$ and $\pi_4 / \left\| (\pi_1, \pi_2, \pi_3)^T \right\|$ the perpendicular distance.

Representing a Plane in 3D

- We therefore say that for a point X to fall on a plane Π, the point must satisfy the condition $\Pi^T X = 0$, or equivalently $X^T \Pi = 0$
- A plane is obviously defined by any three non-collinear points.
  - So given three such points X_1, X_2, and X_3, what is the equation of the plane passing through these three points?
  - The following follows trivially from the previous bullet:
    - $$\begin{bmatrix} X_1^T \\ X_2^T \\ X_3^T \end{bmatrix} \Pi = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$



Representing a Plane in 3D

- $\begin{bmatrix} X_1^T \\ X_2^T \\ X_3^T \end{bmatrix} \Pi = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$
- The above equation indicates that the plane must define the one dimensional null space of the 3×4 matrix formed by the homogeneous coordinates of the 3 points
  - Note: A modern **SVD** based approach can be used to find the null space solution vector Π.
    - You decompose the 3×4 matrix on the left by writing it as $U_{3\times4}\Sigma_{4\times4}V_{3\times4}$.
    - The last eigenvector in V, the one corresponding to the non-zero singular value, is your solution for Π.
    - Please note that we are talking about SVD of an under-determined system ($m < n$). That is when V contains the basis of null space.

The plane at Infinity π_∞

- Let $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$ be the homogeneous coordinate representation of a point in physical $\mathbb{R}^3$.
- We represent a point at infinity by setting $x_4 = 0$. Such points will be called Ideal Points. Obviously, Π_∞ is the set of all ideal points in physical $\mathbb{R}^3$.
- In a manner analogues to the representational of l_∞ in 2D projectivity, we represent Π_∞ by $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$.

The plane at Infinity π_∞

- The very important role played by l_∞ in 2D projectivity is played by the plane at infinity, Π_∞, for the case of 3D projectivity.
- **In physical $\mathbb{R}^3$, all points at infinity lie on a single plane — the plane Π_∞.**

Representing a Line in 3D

- There are several representations (Application Dependent)
  - Span of Two points
  - Span of Two planes
  - Plucker Cooridnates
  - Plucker Matrix

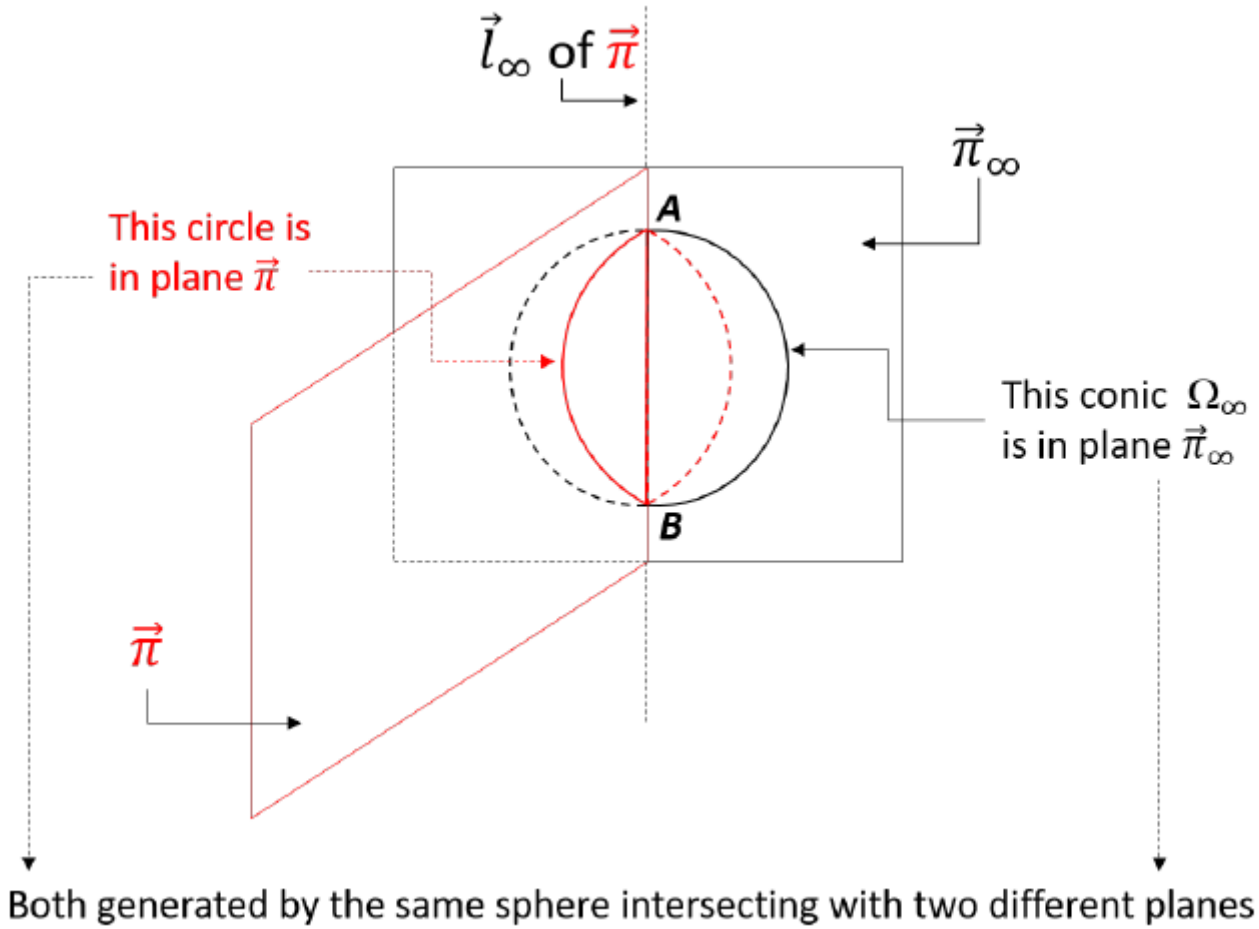
- We will cover them later on when needed.

Representing Quadrics in 3D

- Using homogeneous coordinates, **a quadric** is a surface in 3D that is defined by $X^T QX = 0$, where Q is a 4×4 **symmetric** matrix (**rank =4**) of parameters (in very much the same way that C was a 3×3 symmetric matrix of parameters for a conic in 2D).

The Absolute Conic Ω_∞

- The Absolute Conic is the intersection of any sphere with Π_∞, just as the Circular Points were the intersection of any circle with l_∞.



The Absolute Conic Ω_∞

- Representing a world 3D point by $X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$ in homogeneous coordinates, the general equation of a sphere is given by $x_1^2 + x_2^2 + x_3^2 + dx_1x_4 + ex_2x_4 + fx_3x_4 + gx_4^2 = 0$.
  - Since Π_∞ is defined by $x_4 = 0$, the intersection of any sphere with Π_∞ is given by $x_1^2 + x_2^2 + x_3^2 = 0$, which can be expressed as $\begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} I_{3\times 3} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$ which looks like the equation of a conic. However, don't forget this conic resides in the plane Π_∞. This is our Absolute Conic $\Omega_\infty = I_{3\times 3}$. The conic consists only of imaginary points.

The Absolute Dual Quadric

- The dual of the Absolute Conic Ω_∞ is called the Absolute Dual Quadric and denoted Ω_∞^*. Algebraically, we have $\Omega_\infty^* = \begin{bmatrix} I_{3\times 3} & \vec{0} \\ \vec{0}^T & 0 \end{bmatrix}$.
  - **Geometrically, Ω_∞^* consists of all planes tangent to the point conic Ω_∞.**

PROJECTION GEOMETRY

TRANSFORMATIONS OF WORLD 3-SPACE

Transformations of World 3-Space: Projective

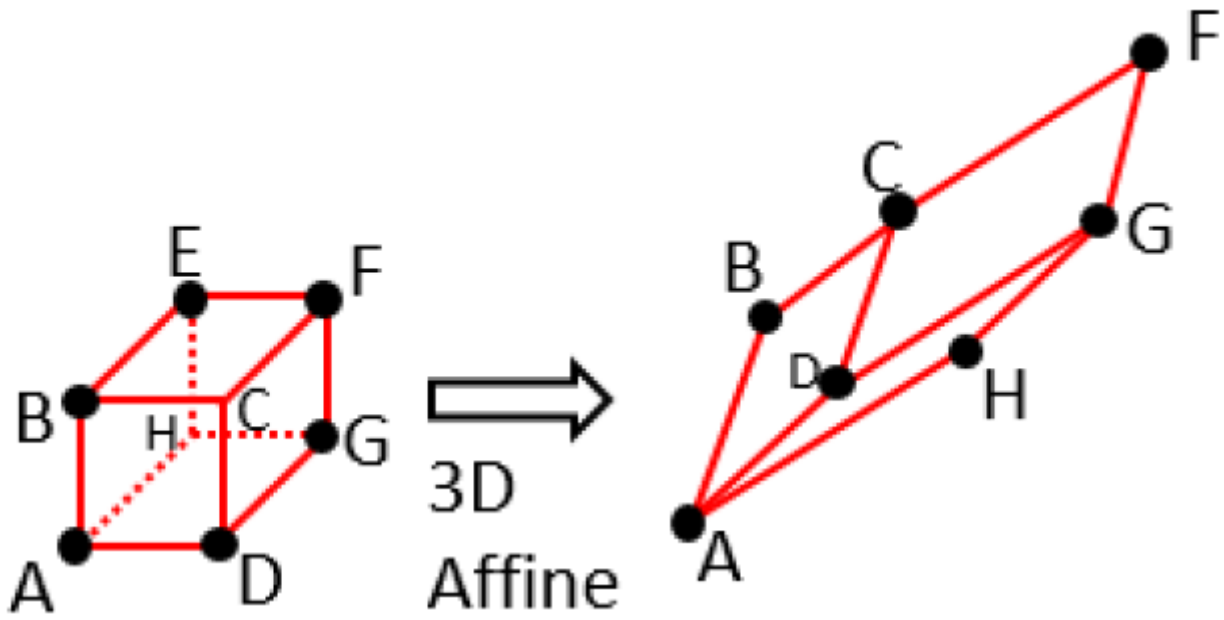
- In homogeneous coordinates, a general transformation of the physical $\mathbb{R}^3$ is achieved with a 4×4 homography: $X' = HX$.
- For the same reasons as for 2D projectivity, a 4×4 homography is best expressed as: $X' = \underbrace{\begin{bmatrix} A_{3\times3} & \vec{t} \\ \vec{v} & v \end{bmatrix}_{4\times4}}_{H} X$
  - Such transformations for all non-singular H form the **Projective Group**.
  - Since only the ratios of the elements of H matter, H can be specified fully with 15 parameters. So, we say that in the Projective Group H has 15 degrees of freedom.

Projective Group maps straight lines into straight lines.

Transformations of World 3-Space: Affine

- The Affine Subgroup of the general Projective Group is specified by restricting the last row of H to $\begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}$.
  - Now we can write $X' = \begin{bmatrix} A_{3\times3} & \vec{t} \\ \vec{0}^T & 1 \end{bmatrix}_{4\times4} X$, and give physical meaning to the 3×3 matrix A and the 3-vector $\vec{t}$.
  - Whereas $\vec{t}$ is simple translation, A represents unequal scaling of space along three orthogonal directions plus rotations.
  - The only requirement on A is that it be non-singular. Since it takes 9 parameters to specify A and 3 to specify $\vec{t}$, 3D affine has 12 degrees of freedom.

Transformations of World 3-Space: Affine



Transformations of World 3-Space: Similarity and Euclidean

- We obtain the Similarity Subgroup of the Affine Group by imposing the additional constraint that $A^T A = \lambda^2 I$, where I is a 3×3 identity matrix.

- similarity transform preserves angles — in other words, it is shape preserving in 3D. That implies that A can only do two things: rotate a 3D object and scale it isotropically.
  - It takes 3 parameters to specify a rigid-body rotation in 3D (two for defining the direction of the axis of rotation and one for the angle of rotation around the axis).
  - In addition we need one parameter for the the isotropic scaling, and 3 for $\vec{t}$. So, in all, a similarity transformation has 7 Dof.
  - We obtain the Euclidean subgroup by further restricting A to $A^T A = I$. That means, A must now be a pure rigid-body rotation. Reasoning as above, we see that an Euclidean transformation has 6 DoF.

Review

Transformation	Matrix	# DoF	Preserves	Icon
translation	$\begin{bmatrix} I \mid t \end{bmatrix}_{3\times4}$	3	orientation	
rigid (Euclidean)	$\begin{bmatrix} R \mid t \end{bmatrix}_{3\times4}$	6	lengths	
similarity	$\begin{bmatrix} sR \mid t \end{bmatrix}_{3\times4}$	7	angles	
affine	$\begin{bmatrix} A \end{bmatrix}_{3\times4}$	12	parallelism	
projective	$\begin{bmatrix} \tilde{H} \end{bmatrix}_{4\times4}$	15	straight lines	

Table 2.2 Hierarchy of 3D coordinate transformations. Each transformation also preserves the properties listed in the rows below it, i.e., similarity preserves not only angles but also parallelism and straight lines. The 3×4 matrices are extended with a fourth $[0^T\ 1]$ row to form a full 4×4 matrix for homogeneous coordinate transformations. The mnemonic icons are drawn in 2D but are meant to suggest transformations occurring in a full 3D cube.

Questions